



Apple-style touch-screen device (Image courtesy: www.mercuryinnovations.com)

Lighter and brighter touch screens—Apple style

Since people spend a lot of time using technology, they like their devices to be more interactive, with more natural forms of engagement. Some like to give voice inputs, while others like gestures. This demand for interactivity is greatly influencing the touch features of displays.

It is not surprising then that Apple adopted in-cell touch sensing in iPhone 5 despite knowing that it would raise the cost of the phone significantly. A normal capacitive touch screen uses a layer of glass on top and a layer of indium tin-oxide (ITO) underneath it. ITO is transparent and electrically-conductive. When a finger is brought near it, it acts like a capacitor—the capacitive change is detected by the circuitry in the phone, and the finger's position tracked. Below the ITO layer is another layer of glass, which has a second layer of ITO on it to isolate the LCD screen from electrical noise. So, actually the image has to go through two layers of glass and ITO. Even if you assume the best-quality glass and minimal refraction, some amount of clarity is lost. Plus, having multiple layers also increases the weight.

In-cell multi-touch technology tries to overcome these problems by combining the LCD and touch sensing

layers. It integrates touch into the thin-film transistor (TFT) LCD manufacturing process, obviating the need for additional sensors and glass.

Apple appears to have been thinking about this since as early as 2006, when it filed a provisional patent for methods of including touch capabilities in displays. In August this year, they were granted a patent for an extensive application that covered several ideas as well as manufacturing

processes that could be used to build the displays. The abstract mentions that there are several forms in which the touch sensing elements may be integrated with the display circuitry: "Touch sensing elements can be completely implemented within the LCD stackup but outside and not between the colour filter plate and the array plate. Alternatively, some touch sensing elements can be between the colour filter and array plates with other touch sensing elements not between the plates."

Japan Display recently showcased three prototype displays in collaboration with Sony, Toshiba and Hitachi. Hyped as 'Innovation Vehicles,' these displays represent the technologies that will be incorporated into products in the recent future. One of the innovative features of these prototypes is the Pixel Eyes technology, which is comparable to in-cell sensing.

According to the company's announcement at FPD International 2012, "In Pixel Eyes technology, the touch-panel is built in, rather than being attached from outside. The structure becomes simple, so it's easy to make the display thin. Such a thin display is very sensitive, so we've utilised that to enable writing with a pen. Currently, finger operation is the norm, but we

would like to provide a pen-drawing solution next."

A screen that can bulge into buttons

We like to swipe and pinch our screens, but there are still times when we wish for a traditional keyboard experience—say, when typing a long document.

Californian company Tactus Technology together with Touch Revolution seems to have an answer. Their new touch panel looks like the normal touch screen displays on existing handheld gadgets but it uses a tactile layer that sits atop the touch screen. This tactile layer is composed of fluid-filled micro-channels, which can alter fluid pressure and redirect the liquid to create blister-like buttons. Similarly, it can also make the buttons disappear in a jiffy.

This morphing technology can be used to give users a keyboard on-demand experience, by transforming a touch screen's flat surface into real physical buttons. While in the initial stages the company has pre-programmed the size and shape of the buttons, in the future it will be possible for the user to configure these too. The tactile layer consumes very little power while forming the buttons, but there is no power consumption once the buttons are active.

This technology is expected to be used in products starting 2013.

Touch magic in large, retail displays

It is not just mobile phones and such personal devices that are into the touch fad. Retail displays are also getting a huge touch makeover. Industry experts claim that users are so used to touch-enabled displays in their personal devices that they expect the same interactivity to be extended to larger-format screens as well. What sets this segment apart is the need for robustness—as the displays will be frequently used by several untrained users.

Spice king McCormick recently opened its World of Flavors—a 353sq.m complex at the Baltimore Inner Harbour, in USA, where it has de-